

NIFTY 50 Event Study: Post-Shock Recovery and a Simple Event-Driven Backtest

Research Note — AlgoChowk

Hypothesis: Following a significant one-day fall in the NIFTY 50, does the index tend to recover over the next few trading days?

1. Objective and Hypothesis

This study tests whether large one-day declines in the NIFTY 50 are followed by positive forward returns. A significant event is defined as a daily close-to-close return of -2% or lower. Because the event is only known after the event-day close, the strategy enters at the next trading day's Open, which avoids using information that would not have been available at entry.

2. Data and Method

Daily NIFTY 50 OHLC data were downloaded from Yahoo Finance using the ^NSEI symbol, covering 4 January 2010 through 22 September 2026. The downloaded sample contains 4,107 rows, with no missing OHLC values and no duplicate dates in the validation output. Forward returns were evaluated over 3, 5, and 10 trading days. A 5 basis-point transaction-cost/slippage assumption (0.0005 per trade) was applied. The analysis also includes a same-sample baseline, a one-sample t-test, an out-of-sample split, threshold robustness checks, and a simple 5-day event-driven backtest with overlapping events skipped.

3. Event-Study Results

Holding	N	Mean	Median	Win rate	Std. dev.
3 days	108	0.2557%	0.4988%	59.26%	3.09%
5 days	108	0.1213%	0.5873%	53.70%	4.23%
10 days	108	0.7150%	1.1813%	59.26%	6.22%

There were 108 qualifying events. The observed event-window means were positive for all three holding periods, with the 10-day window showing the largest mean return in this sample. The win rate was above 50% for each window. These figures describe the sample; they do not by themselves establish that the post-shock effect is statistically reliable or that it would persist in future data.

4. Backtest Results

Metric	Observed result
Trades	80
Cumulative return	16.95%
Maximum drawdown	-27.42%

The simple 5-day event-driven backtest generated 80 non-overlapping trades. The reported cumulative return was 16.95%, while maximum drawdown was -27.42%. This is a historical backtest result under the stated rules and cost assumption, not a forecast of future performance.

5. Robustness, Limitations and Interpretation

The research engine varies the event threshold and keeps the rest of the methodology fixed, reducing the risk of treating a single parameter choice as definitive. Important limitations remain: the event sample is relatively small (108 observations), market regimes can change, historical data may contain vendor-specific issues, and transaction-cost/slippage assumptions are simplified. The backtest also skips overlapping events, which changes the trade sample relative to the raw event count. Most importantly, the event means should be compared with the normal-market baseline and the out-of-sample results before drawing a strong conclusion.

6. Conclusion

Within the tested 2010–2026 sample, NIFTY 50 days meeting the -2% event threshold were followed by positive average returns over 3-, 5-, and 10-trading-day windows. The 5-day event-driven backtest produced a 16.95% cumulative return with a -27.42% maximum drawdown. These results provide evidence relevant to the recovery hypothesis, but they are not sufficient on their own to establish a persistent trading edge. Baseline comparison, statistical testing, out-of-sample behavior, and robustness are necessary for a more complete assessment.

Reproducibility: The accompanying Jupyter notebook contains the data download, event detection, forward-return calculations, baseline comparison, statistical test, robustness analysis, and event-driven backtest.